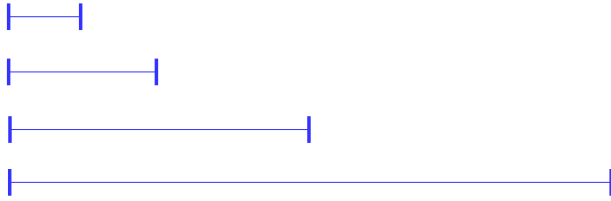


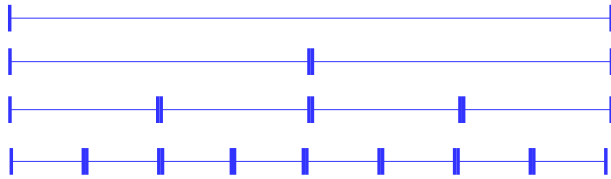
Basis

Data types



```
sint8  uint8  = byte
sint16 uint16
sint32 uint32 = int  uint
sint64 uint64 = slen ulen
```

Registers



R0 - R15 64 bit

R0

R0.32.0-1

R0.16.0-3

R0.8.0-7

Special registers

PC	64 bit	processor counter
LR	64 bit	link register
SP	64 bit	stack pointer
BP	64 bit	base pointer
OP	64 bit	object pointer
CTX	64 bit	context

TBP (ro) 64 bit thread base pointer
CLK (ro) CPU clock counter

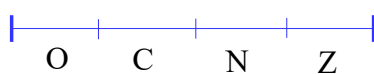
F 4 bit O C N Z

FLAGS 64 bit F.0-15

F.0-7 immediate

F.8-15 sticky

F.0.C bit C in flag block 0



Ordinary commands

Rdst = Csrc

Rdst = unop RCsrc

Rdst = RCsrc1 binop RCsrc2

[Cond] Rdst = RCsrc1 binop RCsrc2

Rdst,F.0 = RCsrc1 binop RCsrc2

jmp RCsrc

jmp PC+Csrc

call RCsrc

call PC+Csrc

ret

Dst = RCsrc

Rdst = Src

```

Dst = *Src
Dst = *(Src+RCsrc)
Dst = *(Src+Csrc*Rsrc)
Dst = *(Src+Csrc*Rsrc+Csrc)

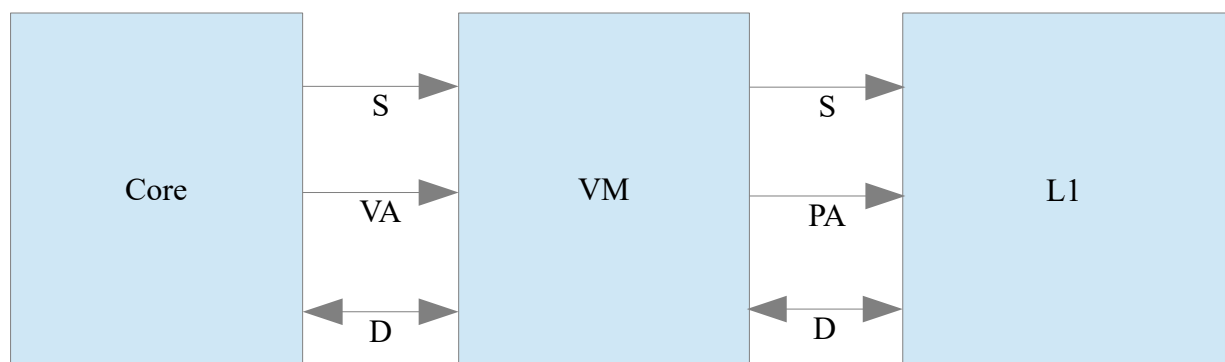
```

```

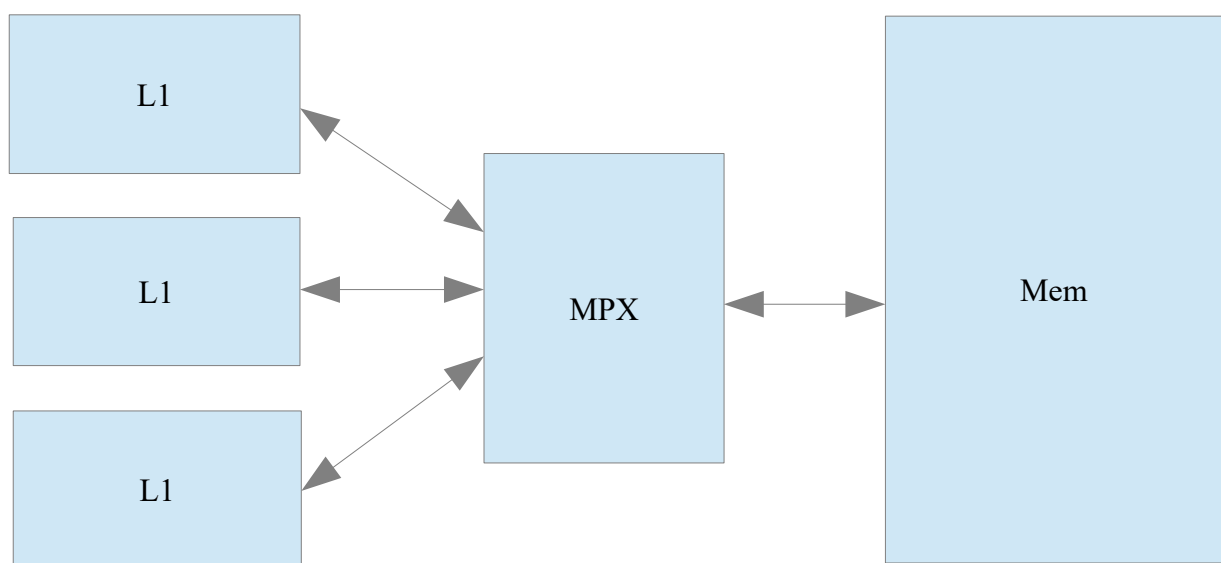
*Dst                = Src
*(Dst+RCdst)        = Src
*(Dst+Cdst*RCdst)    = Src
*(Dst+Cdst*Rdst+Cdst) = Src

```

Memory

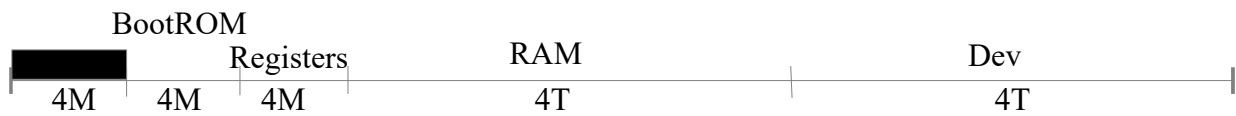


S = Code Data R0Data

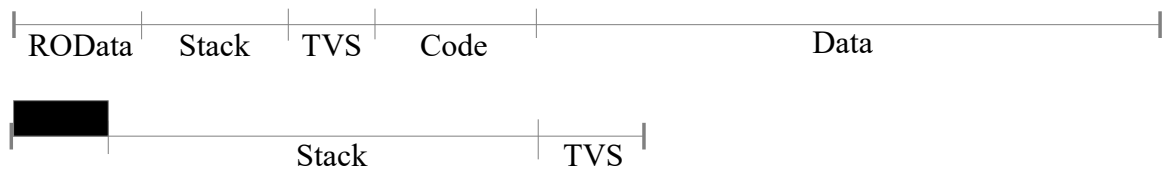


Cache line = 64 byte = 8 Reg

Address space

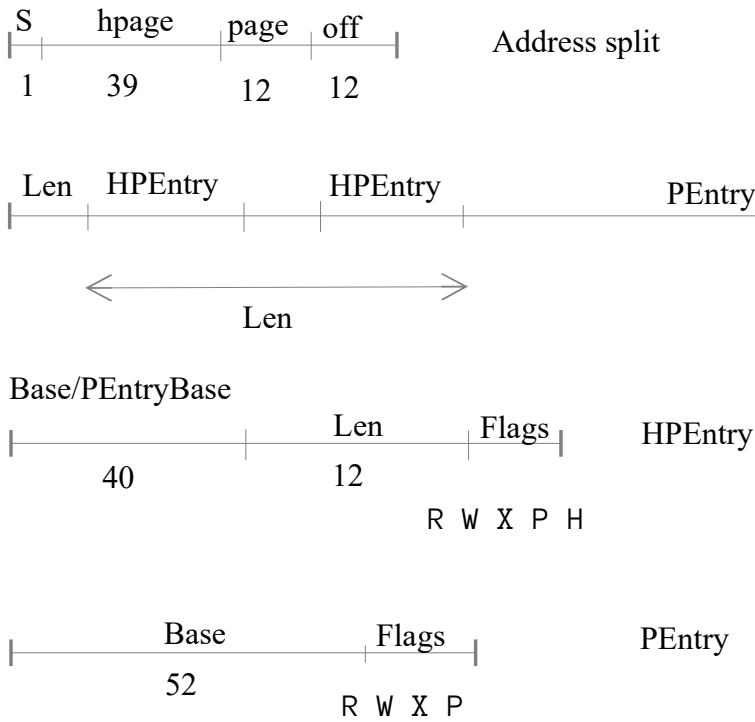


Process



Page aligned, page size = 4K

VMT



Locks

lock address intr disable

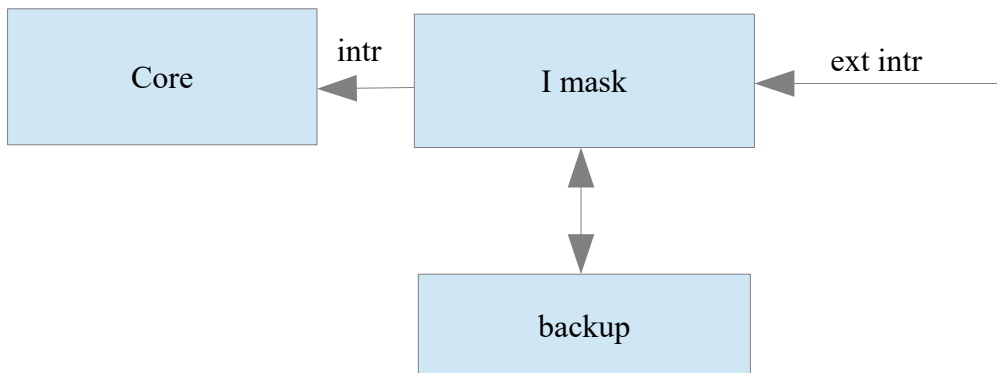
... <= 64 commands

unlock address intr restore

address → 64 bit, 0 OR core number

sysentry intr disable

sysexit intr enable



I mask == 0 in user mode, unless lock active

System registers

VMT 64 bit VMT address

sysVMT 64 bit system VMT address

sysPC 64 bit PC in system mode

sysSP 64 bit SP in system mode

intPC 64 bit PC in interrupt mode

intSP 64 bit SP in interrupt mode

Hidden mode bits

I	1 bit	interrupt mode
S	1 bit	system mode
M	1 bit	RAM enable

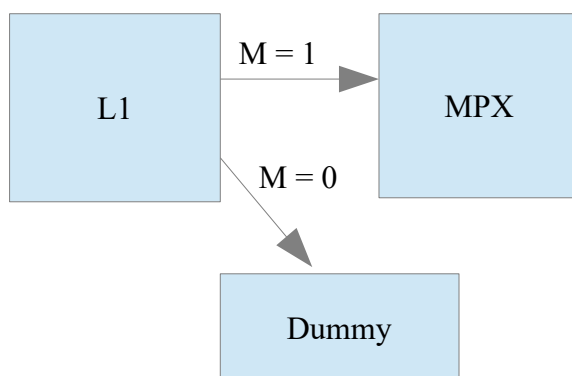
Backup

PC, SP, R0, R1, R2, R3, FLAGS, CLK,

Dual-VM

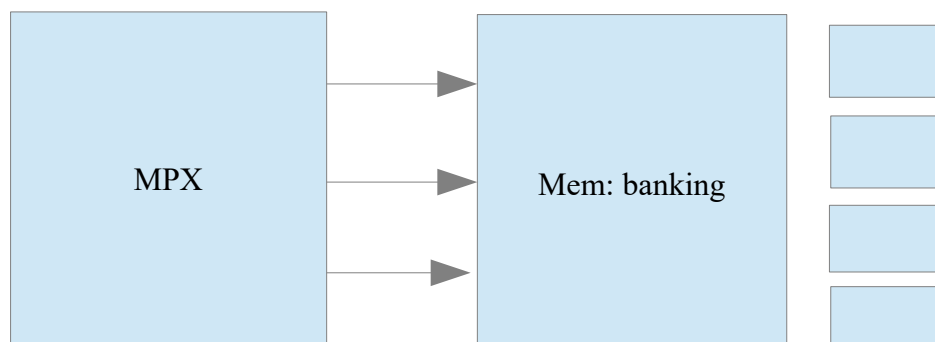


L1 as memory



Dummy: W – ignore R – 0

MPX



Intr controller

Timers

Text display

Keyboard

Disk